Minseok Lee

minseoklee@sogang.ac.kr
https://github.com/ms1ee

Research Interests

Fuzzing, Software Testing, AI for Security

Education

Sogang University — Seoul, Republic of Korea
M.S. in Computer Science and Engineering (Combined B.S.–M.S. Program) — Mar. 2026 – Present
Information Security Lab, Advisor: Prof. Jaeseung Choi

Sogang University — Seoul, Republic of Korea
B.S. in Computer Science and Engineering (CGPA: 3.89/4.3) — Mar. 2020 – Feb. 2026
Graduated ***Magna Cum Laude***

(On leave for 2 years: Mandatory military service)

Research Experience

Graduate Researcher, Information Security Lab — Sogang University
Advisor: Prof. Jaeseung Choi — Mar. 2026 – Present

Directed grey-box fuzzing with Large Language Models

Developing a directed grey-box fuzzer that leverages LLMs to guide fuzzing toward target bugs.

Improving coverage-guided fuzzing

Developed **RareRank**, a frequency-based seed queue minimization technique for rank-aware seed scheduling in grey-box fuzzing. Built on top of AFL++ and evaluated on nine real-world programs, it covers 2.6% and 4.2% more branches than AFL++ and RandSet, respectively, while finding more unique bugs.

Mentored an undergraduate researcher on improving the splicing stage of grey-box fuzzing, contributing to the experiments and the background study.

RareRank (first author) is under review at *IEEE Access*, and the splicing work has been published in *CMC*.

Undergraduate Researcher, Information Security Lab — Sogang University
Advisor: Prof. Jaeseung Choi — Jan. 2025 – Feb. 2026

Built the first version of the **directed grey-box fuzzing with Large Language Models** work above, serving as the sole student researcher on the project, which was commissioned by the National Security Research Institute.

Publications

M. Lee, S. Kim, and J. Choi. "RareRank: Frequency-Based Seed Queue Minimization for Rank-Aware Seed Scheduling in Grey-Box Fuzzing." *IEEE Access*, under review. [artifact]

Y. Ahn, **M. Lee**, S. Kim, and J. Choi. "Enhancing Splicing in Grey-box Fuzzing with Coverage-guided Target Selection and Adaptive Energy Scheduling." *Computers, Materials & Continua (CMC)*, 2026. [paper] [artifact]

Honors and Awards

Combined B.S.–M.S. Program Scholarships, Sogang University Mar. 2025 – Mar. 2026

- Academic stipend of $150/month for 8 months.
- Full tuition (USD $6,180) for the first semester of the M.S. program.

Honor of Magna Cum Laude, Sogang University Feb. 2026

- Awarded to students with a GPA of 3.75/4.3 or higher.

Excellence Award (3rd Place), Capstone Design Competition Dec. 2025

- Ranked 3rd among 25 teams, Sogang University
- Constructed a Korean legal-domain multiple-choice dataset for fine-tuning a large reasoning model, and verified its quality through SFT and RLVR training.
- Presented the work as a poster at the Korea Software Congress (KSC) 2025.

Teaching Experience

Teaching Assistant, Fundamentals of Compiler Construction Fall 2026

Teaching Assistant, Introduction to Computer Systems Spring 2026

Teaching Assistant, Introduction to AI Programming Fall 2024, Fall 2025

Teaching Assistant, Advanced Applied C Programming Spring 2025

Academic Service

Student Volunteer, KIISE SIGPL Summer School 2025 Aug. 2025

- KIISE Special Interest Group on Programming Languages (SIGPL)

Selected Academic Projects

Fundamentals of Compiler Construction | *F#* Fall 2024

- Implemented a simplified compiler with a type checker, AST-to-IR translator, and IR optimization.

Operating System | *C* Fall 2024

- Implemented system calls, process scheduling, and virtual memory in PintOS.

Programming Language | *F#* Spring 2024

- Implemented imperative and functional languages with a type checker.

Skills

Programming: C/C++, Python, F#, OCaml, Assembly Language (x86-64)
Tools: AFL++, LLVM, Git, Docker, GDB, Bash
Languages: Korean (Native), English (TOEFL iBT 5.0/6.0)